Lesson Plan · Wednesday, October 28

Algebra I · Mr. Pandey · Module 2, Topic 2, Lesson 3: Determining Slopes of Perpendicular Lines (Day 1) + Lessons 1–3 Exit Ticket

Objectives (students will…)

- Explain why perpendicular lines have slopes that are negative reciprocals.
- Identify the slopes of horizontal (zero) and vertical (undefined) lines.
- Use the Lesson 1–2 transformation rules: f(x) + d, a·f(x), f(x − h), f(bx).
- Show what they know on the Lessons 1–3 exit ticket.

TEKS

- **A.3E** effects of a·f(x), f(x) + d, f(x − c), f(bx) on f(x) = x
- **A.2E, A.2F** lines parallel / perpendicular through a point **A.2G** horizontal and vertical lines

Materials

- ☐ Bell work: *t2-lesson3-bellwork-day1* (student site)
- ☐ Slides: *t2-lesson3-perpendicular-slopes-catchup* + "Lesson 3 with answers"
- ☐ Student Edition pp. 292, 298, 307–313
- ☐ Lesson 3 video: *videos/T2-Lesson3-Perpendicular-Lines.mp4* (3:24)
- ☐ Exit ticket copies, 2-sided: *T2-Lessons1-3-Exit-Ticket.pdf*
- ☐ Answer key: *T2-Lessons1-3-Exit-Ticket-ANSWER-KEY.pdf*

Agenda (60 minutes)

Time	Activity	What to do / say
0–3	**Bell work** (4 questions)	Reciprocal of ⅔, rotating (8, 5) 90°, slope from two points. Take attendance while they work; go over Q1 only (reciprocal).
3–7	**Lesson 3 video** (3:24) · *T2-Lesson3-Perpendicular-Lines.mp4*	Before it plays: "Listen for what happens to the slope when a line turns 90°." After: students say the rule back: **flip the fraction and change the sign**.
7–20	**Activity 3.1** ★ Slopes of Perpendicular Lines · SE pp. 307–309	Students work in the book in pairs: read the graphs on p. 307, the Worked Example on p. 308, then **Question 4 on p. 309**. Circulate; check that the slopes multiply to −1. `2/3 ⊥ −3/2` `−4 ⊥ 1/4` `1 ⊥ −1`
20–26	**Activity 3.2** ★ Horizontal and Vertical Lines · SE pp. 310–311	Teach **HOY VUX** (Horizontal, 0 slope, y = c; Vertical, Undefined, x = c), then students do **Questions 6–10 on p. 311**. Matches ticket Q3.
26–40	**Book practice like the exit ticket** (pairs, teacher circulates)	One problem from each, in this order: • **Activity 2.1, Question 6 · SE p. 292**: horizontal translations → ticket Q4, Q5 (inside = left/right, backwards) • **Activity 2.3 · SE p. 298**: graph one transformation → ticket Q1, Q6 (a·f(x) and f(bx) change the slope) • **Activity 3.3, Questions 3–4 · SE pp. 312–313**: write the equation of a perpendicular line → ticket Q7, Q8 Last 2 minutes, say the reminders: range = y-values; zero = x-intercept; point-slope from 2 grid points.
40–58	**Exit ticket** (10 questions, independent)	Front and back, 18 minutes. Questions follow End of Topic Assessment items 1–10. IEP/504: read aloud on request, reminders card, extra time (finish at the start of Thursday's class if needed).
58–60	**Wrap up**	Collect tickets. Homework: **Lesson 2 quiz** on the student site. Tomorrow: rest of Activity 3.3 (Question 5) and Talk the Talk.

Check for understanding

- During 3.1, thumbs up/down: "Are slopes 3 and 1/3 perpendicular?" (No, the sign must change.)
- Whiteboards: give the perpendicular slope for 5, −2/7, 0.
- Exit ticket: sort by the key's "next step" line.

After class (using the answer key)

- Missed Q1, 4, 5, 6 → Lesson 1 or 2 video tonight.
- Missed Q7, 8 → reteach in tomorrow's bell work / Activity 3.3.
- Missed Q2, 3, 9, 10 → Topic 1 skills, Mon 11/2 review day.

Changes from the pacing calendar: Getting Started is replaced by the Lesson 3 video. Activity 3.3 Questions 3–4 move up from Thursday so students practice ticket Q7–8 in the book first. The Topic 1 catch-up (L6: write, evaluate and solve a function) moves to Thursday.